

Indian Statistical Institute

BSDS II

Stochastic Process

Class Test I (sample solution)

1. Let p_k be the probability of the pattern of two consecutive Heads (HH) or the pattern of Head followed by Tail (HT) is obtained for the first time after the k -th toss of a coin. Assume that the tosses are independent and the success probability of each toss is $0 < p < 1$. Find a recurrence relation between p_k and p_{k-1} for $k \geq 3$. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.

Sample Solution: We note that $p_0 = p_1 = 0$. For $p_2 = \mathbb{P}(HH) + \mathbb{P}(HT) = p^2 + pq = p$. Now, for $k \geq 3$, we note that if the first toss is a H, then the next toss must result in H or T, i.e., one of the pattern appears after second toss. But we have assumed that $k \geq 3$. So, we must have the first toss as T. For any sequence length k of H and T which results in the first occurrence of either of the the patterns HH or HT, after k tosses, we take tosses from 2 until the k -th toss until the k -th (length of $k - 1$), we will have an occurrence of either of the patterns HH or HT. Conversely, for any sequence length $k - 1$ of H and T which results in the first occurrence of either of the the patterns HH or HT, after $k - 1$ tosses, and put T if front of the sequence, the new sequence (of length k) will result in the first occurrence of either of the patterns HH or HT after k -tosses. Since the tosses are iid, we have, for $k \geq 3$

$$\begin{aligned} p_k &= \mathbb{P}(\text{First toss T, tosses from 2 to } k \text{ results in first occurrence of either HH or HT}) \\ &= qp_{k-1}. \end{aligned}$$

For generating function $P(s) = \sum_{k=0}^{\infty} p_k s^k$, we have

$$\begin{aligned} P(s) &= \sum_{k=0}^{\infty} p_k s^k = ps^2 + \sum_{k=3}^{\infty} p_k s^k \\ &= ps^2 + \sum_{k=3}^{\infty} qp_{k-1} s^k = ps^2 + qs \sum_{k-1=2}^{\infty} p_{k-1} s^{k-1} \\ &= ps^2 + qs \sum_{k=2}^{\infty} p_k s^k = ps^2 + qsP(s). \end{aligned}$$

Thus, we have

$$P(s) = \frac{ps^2}{1 - qs}.$$

Note, $\sum_{k=0}^{\infty} p_k = P(1) = 1$. Now the expected time is given by $P'(1) = 2 + q/p$.

2. Let $\{Z_n\}$ be a branching process with $Z_0 = 1$. Let G be the pgf of the offspring distribution. Assume that $G''(1) < \infty$ and $G'(1) = \mu$. Find $\mathbb{E}(Z_{n+k} | Z_n = i)$ for $k \geq 1$ and $i \geq 0$. Show that $\mathbb{E}(Z_n Z_{n+k}) = \mu^k \mathbb{E}(Z_n^2)$ for $k \geq 1$. Hence find the covariance between Z_n and Z_{n+k} (in terms of μ) for $k \geq 1$.

Sample Solution: Let us denote by $\xi_{k;n}$ the number of offsprings born to the k -th member of the n -th generation of the branching process. We note that $\mathbb{E}(\xi_{k;n}) = \mu$.

We claim that $\mathbb{E}(Z_{n+k} | Z_n = i) = i\mu^k$. We show by induction. For $k = 1$, given $Z_n = i$, we have

$$Z_{n+1} = \sum_{j=1}^i \xi_{j;n}.$$

Thus, we have

$$\mathbb{E}(Z_{n+1} | Z_n = i) = \mathbb{E}\left(\sum_{j=1}^i \xi_{j;n}\right) = \sum_{j=1}^i \mathbb{E}(\xi_{j;n}) = \sum_{j=1}^i \mu = i\mu.$$

Assume that for any $k \geq 1$. Then, given $Z_{n+k} = j$, from the above calculations, we have

$$\mathbb{E}(Z_{n+k+1} | Z_{n+k} = j) = j\mu.$$

Hence, $\mathbb{E}(Z_{n+k+1} | Z_{n+k}) = Z_{n+k}\mu$. Now, we have

$$\begin{aligned} \mathbb{E}(Z_{n+k+1} | Z_n = i) &= \mathbb{E}(\mathbb{E}(Z_{n+k+1} | Z_{n+k}) | Z_n = i) \\ &= \mathbb{E}(Z_{n+k}\mu | Z_n = i) = \mu \mathbb{E}(Z_{n+k} | Z_n = i) = i\mu^{k+1} \end{aligned}$$

using the induction hypothesis.

From the above we conclude that $\mathbb{E}(Z_{n+k} | Z_n) = \mu^k Z_n$. We have

$$\begin{aligned} \mathbb{E}(Z_n Z_{n+k}) &= \mathbb{E}(\mathbb{E}(Z_{n+k} Z_n | Z_n)) \\ &= \mathbb{E}(Z_n \mathbb{E}(Z_{n+k} | Z_n)) \\ &= \mathbb{E}(Z_n \mu^k Z_n) = \mu^k \mathbb{E}(Z_n^2). \end{aligned}$$

Thus, noting that $\mathbb{E}(Z_n) = \mu^n$ (done in class), we have

$$\begin{aligned} \text{Cov}(Z_n, Z_{n+k}) &= \mathbb{E}(Z_{n+k} Z_n) - \mathbb{E}(Z_{n+k}) \mathbb{E}(Z_n) \\ &= \mu^k \mathbb{E}(Z_n^2) - \mu^k \mu^{n+k} = \mu^k [\mathbb{E}(Z_n^2) - (\mu^k)^2] \\ &= \mu^k \text{Var}(Z_n). \end{aligned}$$